NABHONIL SARKAR / NOVEMBER 2026

# Satellite-observed surface differences around Jakarta, 2020–2025

## The question

Jakarta is a coastal megacity on a low-lying delta, where rapid urban development and documented land subsidence affect buildings, drainage and flooding. Published studies identify groundwater extraction, construction loads and natural sediment compaction as likely contributors to the sinking. Understanding what is changing at the surface is useful context for these problems, even when it cannot measure them directly. [8]

A satellite can show that a place looks different without telling us exactly what happened there. A pond, a building site and a flooded field may all produce a different signal from one year to the next. The first task is to establish where those differences occur. Explaining them requires additional evidence.

This study asks: where and when did satellite descriptions of the surface around Jakarta differ most between 2020 and 2025, and how much can we safely infer from those differences? It examines a rectangle on Java's north coast that includes mainland Jakarta, neighbouring areas and coastal water.

The project began with a broader question about sinking ground: would the same amount of ground movement have different implications beneath homes, factories and open land? That question requires dependable measurements of movement and evidence about the people or assets exposed. The project's own radar results could not establish a reliable absolute sinking rate, so I narrowed the study to surface differences that the available data could describe.

## What the study found

Two patterns stood out. The northwest contained 32.2% of the locations with the most extreme 2020–2025 differences, despite containing only 11.1% of all usable locations. For 85.4% of usable locations, 2022–2023 was the largest of the five annual changes. Follow-up checks showed that this annual jump extended across all nine surface classes observed in the study.

These are findings about satellite-derived descriptions. Their physical explanation remains open. Two coastal image pilots lacked the repeated clear observations required to map land–water transitions, and published ground-motion points covered relatively few of the northwest extremes. The report explains both the patterns and the limits on interpreting them.

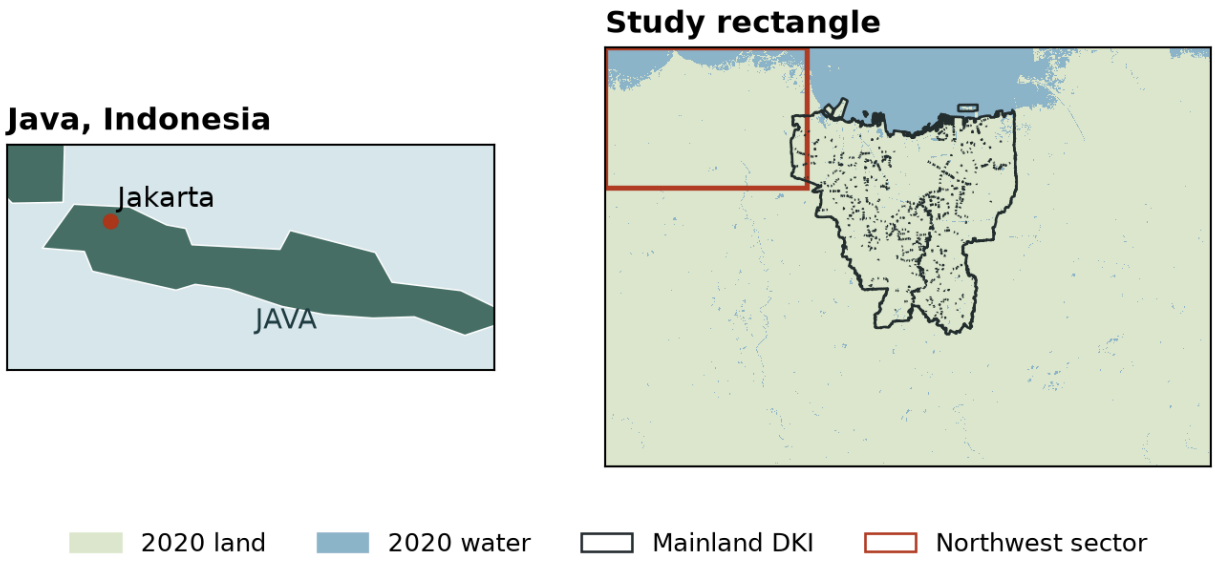



**Figure 1.** The study area and its northwest sector. Blue and pale green show water and land mapped in 2020, not a new coastline survey. DKI is the administrative Jakarta region; the dark outline marks its mainland boundary. Sources: WorldCover 2020 and Natural Earth. [4]

### How I approached it

I compared six annual maps made from AlphaEarth, a Google and Google DeepMind dataset that represents each location using a set of numbers derived from satellite observations. I put the years on the same grid, checked that the processing reproduced an earlier map, and measured how different each location's description was between years. I then tested whether the strongest patterns depended on the map boundary, the type of surface or gaps in supporting observations.

## 01 What satellites can tell us

### Remote sensing means observing from a distance

Remote sensing is the collection of information without visiting every location on the ground. Satellites repeatedly observe large areas, making it possible to compare places and dates consistently. Different instruments observe different properties, and the resulting products answer different questions.

An optical image records reflected light, including wavelengths beyond what our eyes can see. It can help distinguish features such as vegetation, roofs and water. Clouds and their shadows can hide the surface. A radar satellite sends out a signal and measures the return; repeated radar observations can also be used to study ground movement.

### An embedding is a numerical description

AlphaEarth converts satellite information into 64 numbers for each location and year. This set of numbers is called an embedding. The numbers work together as a description: they are not 64 named measurements such as building height, water depth or tree count. A single number should not be read as a physical property. [1]

I compared the whole description between two years. The difference score is called cosine distance. A score near zero means that the two descriptions point in a similar direction in this numerical space; a larger score means they are less similar. It has no units such as metres or millimetres per year. A high score identifies a location to investigate, but does not assign a cause or a type of change.

| INFORMATION | WHAT IT CAN CONTRIBUTE | WHAT IT DOES NOT ESTABLISH ALONE |
|---|---|---|
| **Annual embeddings** | Where and when numerical surface descriptions differ | Construction, land loss or a sinking rate |
| **Dated optical images** | Visible surface conditions at particular times | Persistent change when clear dates are too sparse |
| **Repeated radar observations** | Movement towards or away from the satellite, relative to a reference | Absolute vertical sinking without the necessary checks |

### Surface difference and ground movement are separate

Subsidence means downward movement of the ground. The radar method called InSAR, short for interferometric synthetic aperture radar, compares the phase of returning radar waves across dates. Phase records the position within a wave cycle. Changes in phase can reveal movement along the direction between the satellite and the ground. That direction is called the line of sight; it is not automatically vertical. [2, 3]

A surface can change its satellite description without sinking, and ground can sink while its surface description stays similar. This study keeps those two forms of evidence separate. It also distinguishes a physical hazard from risk, which depends on who or what is exposed and how vulnerable they are.

## 02 Where the study looked and how it compared years

The study covers approximately 85 by 59 kilometres on Java's north coast. Mainland Jakarta lies within the rectangle, but much of the rectangle is outside its administrative boundary. The rectangle was divided into nine sectors to describe broad geographical patterns.

Throughout this report, "northwest" means the upper-left ninth of the study rectangle. It does not mean the administrative area called North Jakarta. In fact, 96.8% of the usable northwest locations lie outside mainland Jakarta. This distinction matters when comparing the results with city-based statistics.

### The comparison followed the same locations through time

First, I sampled all six annual products on one fixed grid with nominal 30 metre spacing. A grid cell is a small map unit, and its centre is the location used for the comparison. There were 5,619,719 usable centres inside the rectangle, including water, with valid descriptions in all six years.

Second, I checked the processing against an earlier 2023 export. Every one of the 64 values matched at every usable centre. This establishes that the chosen sampling procedure reproduces that export; it does not prove that the source product is equally consistent across different years.

Third, I calculated differences for each neighbouring pair of years and directly between 2020 and 2025. The endpoint comparison is not the sum of the five annual differences. I also calculated a direct 2020–2023 comparison. Finally, I tested important alternative explanations and checked the availability of independent evidence.

## 03 The northwest contained a disproportionate share of extremes

To identify the strongest endpoint differences, I ranked the usable locations across the whole rectangle and selected the highest 1%. These are called extreme cells here. “Extreme” is a ranking within this dataset, not a threshold for confirmed physical change or danger.

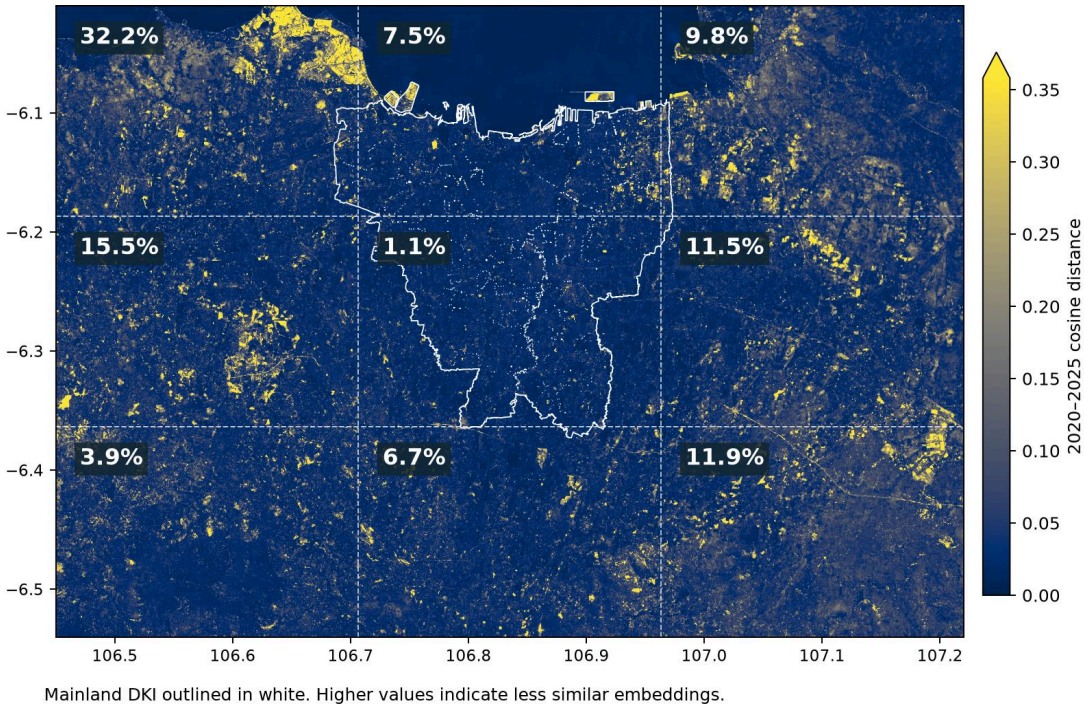



**Figure 2.** Differences between 2020 and 2025. Brighter colours mean less similar annual descriptions. Each percentage is that sector’s share of all extreme cells across the rectangle, not the percentage of the sector that changed. The white outline marks mainland Jakarta. Colours are capped for display; calculations use uncapped scores.

The northwest contained 18,096 of the 56,198 extreme cells: 32.2% of the total, on 11.1% of usable cells. The concentration persisted under two checks, although it weakened when the comparison was restricted to land mapped in 2020.

| COMPARISON | NW SHARE OF USABLE CELLS | NW SHARE OF EXTREME CELLS |
|---|---|---|
| **All usable cells** | 11.1% | 32.2% |
| **2020 land only** | 10.4% | 23.2% |
| **Western source strip excluded** | 10.2% | 33.0% |

The last two rows recalculate the extreme threshold within their retained cells. The land-only check excludes locations mapped as water in 2020, including any that might have become land later. None of these checks identifies reclamation, land loss or development at an individual site.

## 04 The annual maps reveal a widespread jump

For 85.4% of usable locations, 2022–2023 was the largest of the five annual changes. The maps below show why this stands out: the brightening is spread widely across the study rectangle, rather than confined to the northwest. The northwest endpoint concentration and the widespread annual jump answer different questions about place and time.

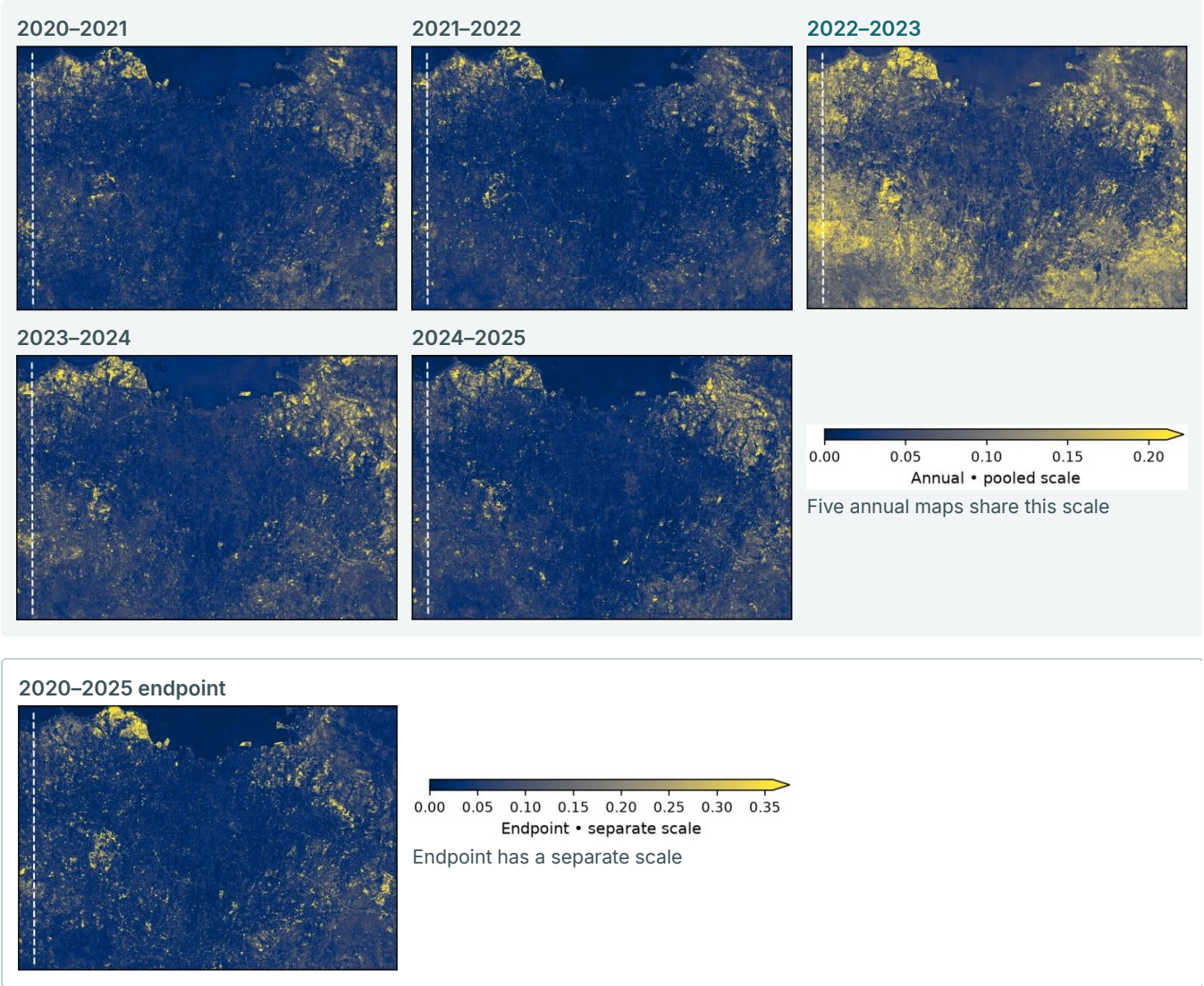



**Figure 3.** The five annual comparisons and the direct 2020–2025 endpoint, using the existing report panels. Brighter colours indicate larger differences. The annual maps share one colour scale; the endpoint uses a different scale, so equal colours should not be compared between them. Dashed lines mark the tested source-tile seam.

### A pattern in the descriptions can have several explanations

In 2024, 81% of locations were closer to their 2023 description than to their 2022 description. In 2025, 68% were closer to 2022 than to 2023. Many descriptions therefore moved back towards the earlier state, but this does not mean that buildings disappeared or land returned to an earlier condition.

Temporary surface conditions, observation conditions and differences in the source product remain possible explanations. A direct 2020–2023 comparison ends in the jump year; the 2020–2025 comparison may be less affected by that particular shift. This does not establish which comparison is more accurate.

## 05 The jump was broad but differed between surfaces

The middle, or median, difference score was 0.069 for 2022–2023, compared with 0.024 to 0.032 for the other annual comparisons. The bar chart makes the timing visible without assigning a physical meaning to the score.

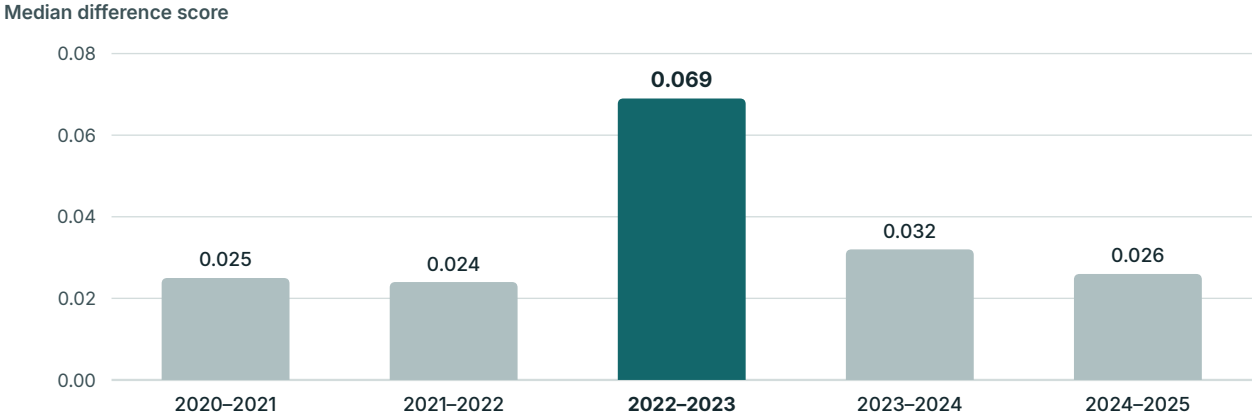



**Figure 4.** Median difference at the same 5,619,719 usable locations in each annual comparison. Values are read from the saved comparison summaries and rounded to three decimals. The score has no physical units.

### Tree cover showed a larger contrast than built surfaces

A first follow-up grouped locations using a separate 2020 map of surface classes, including tree cover, built-up land, cropland and water. All nine classes present had higher median differences in 2022–2023 than in each other annual interval. The same positive pattern appeared in all 64 class–sector combinations with sufficient observations. [4]

The size of the increase was not uniform. For each location, I subtracted its score in another annual interval from its 2022–2023 score, then took the median within each surface class. Repeating this for the four other intervals gives the following ranges of median paired contrasts:

| SURFACE CLASS IN 2020 | RANGE ACROSS THE FOUR COMPARISONS |
|---|---|
| **Tree cover** | +0.058 to +0.065 |
| **Built-up land** | +0.016 to +0.024 |
| **Permanent water** | +0.012 to +0.024 |

The contrast was substantially larger for tree cover than for built-up land. These ranges describe four different comparisons; they are not uncertainty intervals. Nor do they measure how much vegetation or construction physically changed.

The breadth of the jump is compatible with a shared observation or product influence, but does not establish its cause. The 2020 labels do not prove that a location stayed unchanged afterwards. Geography also differs between classes, so these pooled contrasts are descriptive clues rather than an isolated effect of surface type.

## 06 What other evidence could add

### Could clear images resolve coastal land and water changes?

A coastal imaging test tried two fixed designs across six northwest test areas: one with January–April and June–September seasons, followed by a separate June–September-only design. Clear views were too sparse through time to meet the rules for every area. The first stopped after eight dates in January–April 2020. The second stopped after eight dates in June–September 2022, when four of six areas, including both controls, failed. No land–water mapping was possible under those rules, and no transitions were classified. Both designs used two preselected scenes per month; every available scene or a longer season was not tested. The detailed coverage table is in the notes. [6]

### Published motion points miss many northwest extremes

The northwest's biggest differences fall near published motion points at about half the rate of northwest locations generally. For example, a check using 75 metre square neighbourhoods gives the comparison below. Alternative shapes give the same broad sampling limitation; their figures are in the notes.

| ILLUSTRATIVE SAMPLING CONVENTION | ALL USABLE NW CELLS INSIDE | EXTREME NW CELLS INSIDE |
|---|---|---|
| **75 m square neighbourhoods** | 20.14% | 9.12% |

Being inside one of these neighbourhoods does not mean that the surface was directly measured there. The result limits how well the published 2017–2023 motion data could explain the northwest's 2020–2025 differences; it does not reveal why observations are missing. The planned comparison remains on hold; no motion values have been aggregated or compared with embedding differences. [5, 7]

### The original radar work retained only a relative pattern

The project's own InSAR results had a separate limitation. One viewing track had no area passing the strict quality screen in North Jakarta. The other retained a relative pattern over 68.4% of North Jakarta, but its median rate shifted by about 8.05 millimetres per year when the reference area changed, beyond the preset limit of 3 millimetres per year. The reference supplies the zero of the measurement. I retained the relative pattern and withdrew the absolute-rate claim.

## 07 What the project establishes and what remains open

### A repeatable description of where and when differences occur

The completed contribution is a consistent six-year comparison, two clear descriptive patterns and a record of checks that constrain their interpretation. The northwest concentration survives the land-only and western-strip checks. The widespread 2022–2023 jump extends across the observed baseline surface classes. These results identify questions and locations that merit explanation.

The processing check is strong within its purpose: all 64 values of the earlier 2023 export were reproduced at all 5.6 million usable grid centres. That does not make every mapped difference a physical change, establish accuracy across years or make neighbouring cells independent observations.

### Independent interpretation remains limited

The project did not establish whether the largest differences represent construction, pond changes, temporary flooding, vegetation conditions or product effects. The optical pilots could not meet their observation rules. The published motion sampling misses much of the northwest extreme set.

A separate attempt to compare embeddings with simpler vegetation and built-up indicators also stopped before scoring. Across 93 releases of an imagery archive, candidate images at both endpoints were available in only 16 geographical blocks reserved for developing the comparison, below the 20 required. Catalogue listings for higher-resolution imagery did not provide access to the images themselves. There is therefore no demonstrated performance advantage over those simpler indicators.

The original risk-zoning goal remains unmet. Reliable hazard measurements and adequate analysis of exposed people or assets were not established. The difference maps should not be used to designate dangerous neighbourhoods, estimate sinking rates or identify confirmed land loss or reclamation.

### The next question is what kind of change occurred

If further work is chosen, it should ask what kinds of surface change produced the differences. Urban expansion, pond changes, vegetation changes and temporary water conditions are possible subjects to investigate, not explanations established by this study. A useful comparison would inspect sites selected by a fixed rule, including ordinary-score sites as well as extreme ones, using dated images and consistent labels.

The tree-cover contrast concerns the 2022–2023 annual jump and does not establish its cause. Whether urban expansion accounts for the 2020–2025 endpoint extremes remains untested. Before committing to a larger study, the availability and interpretability of endpoint images would need to be established. No new attribution study or motion comparison is presented here.

### The answer to the study question

The strongest endpoint differences are disproportionately concentrated in the northwest of the study rectangle, while the strongest annual jump occurs widely in 2022–2023. The study can describe those patterns and show which checks they survive. It cannot yet tell us what physical processes produced them.

NOTES

# Supporting observation notes

### Clear observations in the two coastal designs

All six one square kilometre windows lie in the northwest and were selected geometrically using WorldCover 2020, not the embedding difference map. C1–C4 are coastal windows; L1 is an inland control and W1 a water control. Both designs used two dates per month on a separate 20 metre grid.

A cell needed at least three clear dates across at least three months, a span of at least 60 days and at least three dates separated by ten days or more. At least 80% of cells in every window had to qualify under both buffers. These 60 and 120 metre buffers enlarge the excluded areas around unsuitable observations. Temporal support means passing all these observation conditions.

| WINDOW | JAN–APR 2020 60 M | JAN–APR 2020 120 M | JUN–SEP 2022 60 M | JUN–SEP 2022 120 M |
|---|---|---|---|---|
| **C1** | 0.00% | 0.00% | 0.00% | 0.00% |
| **C2** | 0.00% | 0.00% | 93.12% | 90.24% |
| **C3** | 0.00% | 0.00% | 58.72% | 57.84% |
| **C4** | 0.00% | 0.00% | 95.76% | 94.76% |
| **L1** | 0.08% | 0.00% | 51.08% | 33.28% |
| **W1** | 0.00% | 0.00% | 21.88% | 21.88% |

**Table 1.** Percentage meeting every support condition. Each entry needed to reach 80%. Later seasons were not acquired and their support is unknown. The optical branch is closed under the tested scope. [6]

### Overlap with published motion sampling

After screening and excluding 40,661 coordinates with conflicting records, 220,625 eligible published locations remained in the rectangle, including 48,883 in mainland Jakarta. The overlap check below holds the qualifying set of one kilometre units fixed. A square or disk is a provisional neighbourhood, not a known measurement footprint. An occupied 250 metre subregion contains at least one eligible point.

| SAMPLING REGION | ALL USABLE NW CELLS INSIDE | EXTREME NW CELLS INSIDE |
|---|---|---|
| **75 m square neighbourhoods** | 20.14% | 9.12% |
| **75 m diameter disks** | 15.93% | 7.21% |
| **Occupied 250 m subregions** | 56.77% | 29.26% |

**Table 2.** Percentages use all usable northwest cells or all northwest extreme cells as their respective denominators. Overlapping regions count each embedding centre once; broader overlap does not create measured coverage. [7]

The screen retains 3,459 full one kilometre units with at least ten eligible points, at least four occupied 250 metre subregions and points in both halves of each coordinate axis. Of these units, 413 intersect the northwest and are not necessarily wholly inside it. The published nominal 75 metre scale does not establish an exact footprint or positional-error bound.

SOURCES

# Methods and sources

**Grid and ranking.** The rectangle spans 106.45–107.22°E and 6.54–6.01°S. AlphaEarth's stored 20 metre overview is sampled onto the frozen nominal 30 metre grid. Encoded values are decoded and all 64 components compared. The endpoint extremes use the rectangle-wide 99th percentile with ties included; northwest extremes are a subset of that same ranking. Map cells are not independent ground measurements.

**Map boundary check.** A test fixed before examining contrasts compared the source-tile seam with nearby parallel lines. Neither primary comparison, 2022–2023 or 2020–2025, flagged an unusually sharp step at the seam. All six water-only tests were inconclusive because too few segments qualified. The test cannot rule out a shift affecting both tiles together.

**Records and access.** The public repository contains methods, code, the reports and summaries of completed checks. Detailed sampling tables, source rasters and local evidence bundles are held locally and are not included in a fresh clone. The surface-class contrasts and annual chart reproduce existing saved summaries; no new scientific analysis is introduced here.


[1] Brown and colleagues 2025 — AlphaEarth Foundations

AlphaEarth annual dataset and sampling guide

[2] Lazecký and colleagues 2020 — LiCSAR

[3] Morishita and colleagues 2020 — LiCSBAS

[4] Zanaga and colleagues 2021 — ESA WorldCover 2020

[5] Ohenhen and colleagues 2026 — Land subsidence on Java Island

Published Java motion data

[6] Sarkar 2026 — Optical feasibility supplement

[7] Project methods and code — detailed diagnostic records held locally

[8] Abidin, Andreas, Gumilar and Brinkman (2015). Study on the risk and impacts of land subsidence in Jakarta. Proc. IAHS, 372, 115–120.



The AlphaEarth Foundations Satellite Embedding dataset is produced by Google and Google DeepMind. WorldCover: © ESA WorldCover project 2020, v100, CC BY 4.0. Regional inset: Natural Earth, public domain. Optical follow-up: Copernicus Sentinel-2 data. Published Java data: CC BY 4.0. Third-party terms are recorded in the repository.